

Pharmacological VCP inhibition confers neuroprotection in the *Prph2^{Rd2}* model of retinal degeneration.

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Objective: Inherited Retinal Degenerations are genetically heterogeneous disorders leading to progressive photoreceptor dysfunction and vision loss. Despite their genetic diversity, shared pathogenic mechanisms converge on disrupted proteostasis. Pharmacological inhibition of valosin-containing protein (VCP), a central regulator of protein homeostasis, has previously demonstrated neuroprotective effects in rhodopsin-associated autosomal dominant models as well as in phosphodiesterase 6 (PDE6)-associated autosomal recessive models of retinal degeneration. Here, we evaluated the neuroprotective potential of UPCDC-30766, a recently developed VCP inhibitor, in the *Prph2^{Rd2}* mouse model, which exhibits structural outer segment defects and progressive degeneration affecting both rod and cone photoreceptors.

Methods: Retinal explants from *Prph2^{Rd2}* mice were cultured using an interphase system from PN12 to PN18, corresponding to the peak of retinal degeneration in this model. This ex vivo system preserves retinal lamination and cellular architecture while allowing controlled pharmacological intervention during the degenerative phase. Cultures were maintained in a defined, serum-free medium to enhance reproducibility and minimize experimental variability. Explants were treated with UPCDC-30766 in escalating doses, and DMSO-treated samples served as vehicle controls. Neuroprotection was assessed by ONL thickness and photoreceptor cell rows, TUNEL-positive cell quantification within the ONL, OS length, and the ONL-to-OS ratio of rhodopsin fluorescence intensity to evaluate alterations in rhodopsin trafficking and localization.

Results: Treatment with UPCDC-30766 resulted in significant preservation of photoreceptor row numbers across all tested doses compared to vehicle controls. Although the percentage of TUNEL-positive cells per section did not reach statistical significance, a consistent reduction in apoptotic profiles was observed in treated explants, indicating a trend toward decreased photoreceptor cell death.

Conclusion: UPCDC-30766-mediated VCP inhibition confers structural neuroprotection in *Prph2^{Rd2}* retinal explants. These findings highlight the value of organotypic interphase cultures as a controlled platform for mechanistic studies and preclinical pharmacological testing in genetically diverse retinal degenerations.